Problem 13: Graph Neural Network Platform for Molecular Property Prediction and Drug-Candidate Screening

Domain: Computational Biology | Graph ML | Cheminformatics | Uncertainty

Difficulty: Hard

Status: NEW (added to reach 15)

Problem

Screening millions of molecules in the lab is slow and expensive. ML can prioritize candidates by predicting properties like toxicity, solubility and activity, but models often fail on new chemical scaffolds and give overconfident predictions.

Objective

Build a platform that predicts multiple molecular properties from structure using graph neural networks, estimates uncertainty, explains predictions, and ranks candidates for screening.

Users

Medicinal chemists, computational biology researchers.

Core Requirements

- Accept SMILES strings or molecule files.
- Validate and standardize molecules (RDKit).
- Featurize as molecular graphs (atoms, bonds).
- Train a GNN (GCN/GIN/MPNN/AttentiveFP) on at least two property tasks.
- Baselines: fingerprints (ECFP) + random forest/XGBoost.
- Scaffold-based train/validation/test split.
- Uncertainty estimates (ensembles, MC-dropout or conformal).
- Explanation: highlight atoms/substructures driving the prediction.
- Rank and filter candidates by predicted property, uncertainty and drug-likeness (Lipinski, QED, SA score).
- Web app for batch screening with downloadable results.

Advanced Challenges

- Out-of-distribution scaffold generalization.
- Multi-task learning with missing labels.
- Pretraining (self-supervised on unlabeled molecules).
- Calibration under distribution shift.
- Active learning: choose molecules to “label” next.
- Optional generative extension (propose analogs, then re-score).

Suggested Datasets

MoleculeNet (Tox21, BBBP, ESOL, FreeSolv, HIV), Therapeutics Data Commons (TDC), ChEMBL, ZINC for unlabeled pretraining.

Suggested Tech

Python, RDKit, PyTorch Geometric or DGL-LifeSci, scikit-learn, XGBoost, GNNExplainer/ Captum, Streamlit, FastAPI.

Reference Architecture

SMILES → RDKit cleaner → graph featurizer → GNN encoder (optionally pretrained) → multi-task heads → ensemble/conformal layer → explainer → ranking + filters → UI.

Evaluation

ROC-AUC / PR-AUC (classification), RMSE (regression), scaffold-split vs random-split gap, calibration (ECE), conformal coverage, enrichment factor in top-k, comparison to fingerprint baselines.

Safety Note

For research on benign therapeutic properties. Do not build or release tools aimed at designing

toxic agents.

Pitfalls

Random splits leaking near-duplicate molecules; ignoring label noise in Tox21; claiming biological efficacy from in-silico results.

Suggested Timeline (48h)

0-8h data + RDKit pipeline + baselines; 8-20h GNN training; 20-30h scaffold eval + uncertainty; 30-40h explanations + ranking; 40-48h app.

Standard Judging Rubric (applies to every problem)

Problem understanding	10%
Technical depth	20%
Data engineering (quality, leakage prevention, reproducibility)	10%
Model performance (metrics, baselines, honest evaluation)	15%
System architecture	15%
Working prototype	10%
Explainability and responsible AI	10%
Innovation and scalability	5%
Presentation and demo	5%

Mandatory Submission Checklist

- □ GitHub repository with setup instructions
- □ Working prototype or recorded demo
- □ Documented system architecture
- □ Dataset sources and preprocessing methodology
- □ Evaluation results with baseline comparisons
- □ Limitations and failure cases
- □ One technically challenging feature demonstrated
- □ Short presentation: problem, approach, results, future work